Francisco de Assis de Souza Rodrigues

Computer Scientist · Software Engineer AI · LLMs · NLP · Backend

PhD candidate and MSc in Computer Science (CIn / UFPE). Co-founder and engineer at Octor. NLP publication at LREC 2020 (DeepNLPF).

Full curriculum: Career path

Email · LinkedIn · GitHub · Lattes · ORCID

Skills

AI / NLP: LLMs (vLLM, tool agents) · Transformers · NLI / typed decisions (Laya, MiniLM) · ASR (Whisper) · OCR · RAG · text mining · information extraction · Learning Analytics

Backend: Python · Rust (Actix) · FastAPI · Redis · microservices · REST APIs · Docker · MySQL / PostgreSQL / SQLite · AWS · Linux · OpenAPI

Relevant experience

Octor — Co-founder, Computer Scientist & Software Engineer · present

octor.com.br — clinic management. Production AI and backend:

- Clinical LLM agent (**Sereia**): Rust + Redis + GPU (Whisper, vLLM) with clinical tools
- ASR transcription, typed-decision engine (`platform-decide` / NLI), Laya, WhatsApp SDR, OCR/extraction
- Microservices and APIs (scheduling, EHR, billing, integrations)

NeuralMind — AI / NLP · 2022

NeuralSearchX: neural search, Transformers, QA, FastAPI, AWS/Azure · FACEPE grant

NoVerde / Dotz — Backend Python · 2023

Credit APIs, tests, Ruby→Python migration, AWS Lambda/SQS/S3, Airflow, ELK

Realm · Zafie · CoWatch · CreateLab

Founder/co-founder: mobile DOOH (BagLed), digital menus (food), agritech (herd), education/robotics — details in Career path

Teaching and extension

UniFAP — Former lecturer in **ADS** and **Information Systems** and Coordinator of NExTI (student projects for companies, workshops, NExTI AI). Courses: Biohacking/Deep Web/Cryptography, Data Science (EAD), Mobile (IS), Software Engineering, Internship, Computer Organization & Architecture, Capstones (IT Governance; Web), Special Topics in IS. Also: dozens of thesis boards, project evaluation, and academic event organization — full detail in Career path.

Education

- **PhD** Computer Science — UFPE/CIn (in progress, CAPES) · AI / Learning Analytics & assistive technology
- **MSc** Computer Science — UFPE/CIn (2017–2019, CAPES) · DeepNLPF · LREC 2020
- **Systems Analysis** — UniLeão (2011–2014) · DAVI (assistive tech)
- **MBA** Higher Education Teaching — IDJ (2015–2016)

Links

- Career path (full CV) · Publications · Projects · Startups